



Vishal S R 250701844-2025-CSE ▾

V2**Started on** Friday, 21 August 2026, 8:58 AM**State** Finished**Completed on** Monday, 24 August 2026, 10:36 PM**Time taken** 3 days 13 hours**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2  int max(int high,int low,int arr[],int x,int ans){
3      if(low>high){
4          return ans;
5      }
6
7      int mid=(high+low)/2;
8      if(arr[mid]==x){
9          return x;
10     }
11     else if(arr[mid]<x){
12         ans=arr[mid];
13         return max(high,mid+1,arr,x,ans);
14     }
15     else{
16         return max(mid-1,low,arr,x,ans);
17     }
18
19
20 }
21
22 int main(){
23     int n;scanf("%d",&n);
24     int arr[n];
25     for(int i=0;i<n;i++){
26         scanf("%d",&arr[i]);
27     }
28     int maxnum;
29     scanf("%d",&maxnum);
30     int ans=-1;
31     int result=max(n-1,0,arr,maxnum,ans);
32     printf("%d",result);
33
34 }
```

	Input	Expected	Got	
✓	6 1 2 8 10 12 19 5	2	2	✓

	Input	Expected	Got	
✓	5 10 22 85 108 129 100	85	85	✓
✓	7 3 5 7 9 11 13 15 10	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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